Executive Brief: California Housing Predictive Engine

To: Project Stakeholders
From: Kingsley Egei
Date: March 17, 2026
Subject: High-Impact Predictive Modeling for California Real Estate Values

Robust Market Prediction

This project successfully developed a **Random Forest predictive engine** capable of estimating California median house values with a stable error margin of approximately **$48,310**. The model demonstrated exceptional reliability, with a performance "Delta" showing that it generalizes effectively to new, unseen data which is a critical requirement for any scalable data platform.

Core Findings & Strategic Insights

- **Primary Value Drivers:** My analysis identified **Median Income** and **Geography** (Latitude/Longitude) as the dominant predictors of property value.
- **Proximity Premium:** Data visualization confirmed that **"ISLAND"** and **"NEAR OCEAN"** properties command significantly higher price ceilings compared to **"INLAND"** locations.
- **Model Success:** The model achieved a **"Success" status** because the Test Error (RMSE) did not spike above the Training Error, proving it is not overfit and can accurately interpret real-world housing patterns.

Technical Infrastructure

To achieve these results, I engineered a high-signal feature vector comprising 13 standardized variables:

- **Data Cleaning:** Resolved 207 missing data points via median imputation to preserve sample integrity.
- **Feature Engineering:** Transformed raw block-level aggregates into more meaningful **per-household ratios** (e.g., mean rooms per home).
- **Standardization:** Normalized all features to a mean of 0 and standard deviation of 1 to ensure high-impact variables like income were not mathematically outweighed by larger raw numbers like population.

Conclusion

This model provides a reliable mathematical foundation for automated property valuation. By prioritizing **economic and spatial data**, we can scale this platform to provide high-accuracy insights across the 20,000+ census block groups in California.

Final Project Report

Analysis completed by Kingsley Egei

Status of Report: *Completed 03.17.26*

Project summary and overall findings

Strategic Insight

My analysis identified **Median Income** and **Geography** (longitude/latitude) as the primary drivers of the response variable. By removing redundant raw totals and focusing on engineered means, I created a high-signal feature vector that allows the model to predict house values within a consistent error margin.

Project overview:

This report is designed for anyone learning about data to become productive with their new data science skills by working through a real-life problem using a *California Housing* data set ("housing.csv"). I will use the data set to **predict *median house values***.

The **goal** for this project is for the reader to gain experience in trying out the principles of data science using R as discussed in class.

Each task was outlined in the "**Class Project**" brief. Its findings will be documented below.

The **hypothesis** is that myself as the reader will use his new data science skills to run a model that will predict a valid median house value within an acceptable Test RMSE range of 10-20% within the Training RMSE. Within the 10-20% will validate whether or not the predicted median house value is acceptable and the model is reasonable.

Description of the Data:

This is the California Housing dataset, a classic machine learning dataset commonly used for regression tasks. This data set appeared in a 1997 paper titled Sparse Spatial Autoregressions by Pace, R. Kelley and Ronald Barry, published in the *Statistics and Probability Letters* journal. The researchers built it using the 1990 California census data. It contains one row per census block group.

A block group is the smallest geographical unit for which the U.S. Census Bureau publishes sample data (a block group typically has a population of 600 to 3,000 people). The variables found in the data set are as follows (the names are fairly self-explanatory, and all variables are atomic class numeric with the exception of ocean_proximity which is a factor):

Here's a full breakdown:

Overview

- **20,640 rows** (one per census block group in California)
- **10 columns** (9 numeric, 1 categorical)
- **207 missing values** in total_bedrooms only

Variables

Variable	Type	Description
longitude	Float	Geographic longitude of the block group (range: –124.35 to –114.31)
latitude	Float	Geographic latitude (range: 32.54 to 41.95) — together with longitude, pinpoints location in California
housing_median_age	Float	Median age of houses in the block group (52 unique values; capped at some upper bound)
total_rooms	Float	Total number of rooms across all homes in the block group
total_bedrooms	Float	Total number of bedrooms across all homes — **has 207 missing values**
population	Float	Total population of the block group
households	Float	Number of households in the block group
median_income	Float	Median household income (scaled, not in raw dollars; range: ~0.5 to 15)

median_house_value	Float	**Target variable** — median house value in USD (range: $15K–$500K, capped at $500,001)
ocean_proximity	Categorical	Distance category to the ocean: <1H OCEAN (9,136), INLAND (6,551), NEAR OCEAN (2,658), NEAR BAY (2,290), ISLAND (5)

A few things to note:

- median_house_value is the typical prediction target for regression models.
- total_rooms, total_bedrooms, population, and households are **block-level aggregates**, so it's common to engineer per-household ratios (e.g., rooms per household) for modeling.
- median_income is the strongest single predictor of house value in this dataset.
- The ISLAND category in ocean_proximity has only 5 observations — essentially an outlier class.
- The $500,001 cap on median_house_value suggests values above $500K were truncated, which can affect model performance at the high end.
- Each row pertains to a group of houses representing medians for groups of houses in close proximity.

Description of My Data Pipeline

After walking through the class project requirements, there were multiple steps I needed to take to get to the final product. The following steps were to get the data in the right form, standardized, and cleaned in order to run the final model. My data cleaning process is as follows;

1. Handling Missing Values (Imputation)

Early on I identified 207 `NA` values in the `total_bedrooms` column.

- **The Action:** I calculated the **median** of the existing values (`na.rm = TRUE`) and used it to fill in the gaps.
- **The Goal:** This ensures my Random Forest model doesn't drop those 207 records, preserving my sample size while avoiding the bias that "0" or "Mean" imputation might introduce in skewed data.

2. Categorical Encoding (One-Hot Encoding)

The ocean_proximity variable was a text-based factor (e.g., "NEAR BAY", "INLAND").

- **The Action:** I used model.matrix() to create **dummy variables** (binary 0/1 columns) for each category.
- **The Goal:** Most machine learning algorithms require numeric inputs. By creating separate columns for each location type, I allowed the model to calculate the specific "premium" or "discount" associated with living near the water.

3. Feature Engineering (Ratio Creation)

I derived new insights by combining existing columns.

- **The Action:** I created mean_bedrooms and mean_rooms by dividing the totals by the number of households.
- **The Goal:** The raw number of rooms in a whole district isn't as predictive as the **average size of a home** in that district. This "munging" step turned bulk data into a more meaningful metric for house value.

4. Handling Mathematical Anomalies

Whenever I perform division (like in the step above), I risk creating "Infinite" values or "Not a Number" errors if a denominator is zero.

- **The Action:** I used a logical check (is.infinite and is.nan) to find these errors and reset them to 0.
- **The Goal:** This prevents my scaling or modeling functions from crashing due to non-finite numbers.

5. Feature Scaling (Standardization)

My numeric variables (like longitude and median_income) were on completely different scales.

- **The Action:** I used the scale() function to standardize my predictors so they have a **mean of 0** and a **standard deviation of 1**.
- **The Goal:** While Random Forest is generally robust to scaling, standardizing is a "best practice" career habit. It ensures that variables with larger raw numbers (like population) don't accidentally "outweight" more important variables with smaller numbers (like median_income).

6. Dimensionality Reduction (Column Removal)

- **The Action:** After creating my new "mean" variables and dummy columns, I used `NULL` assignments to remove the original `total_rooms`, `total_bedrooms`, and the original `ocean_proximity` string.
- **The Goal:** This removes **multicollinearity** (redundant data) and ensures my `train_x` only contains the specific features I want the model to learn from.

My final `cleaned_housing` object was a perfectly organized matrix of 14 columns, ready for the `sample()` function and my 70/30 split.

Description of the Statistical Model

Random Forest is a robust, non-linear ensemble model that powers my predictive "blueprint". It was the model the professor recommended us to use for this project.

The Statistical Model: Random Forest

The Random Forest algorithm is an "ensemble" method that constructs a multitude of decision trees during training. Unlike a single decision tree which can be prone to overfitting, the Random Forest averages the results of **500 individual trees** (as specified in my `ntree = 500` parameter) to improve predictive accuracy and control error variance.

- **Recursive Partitioning**: Each tree in my forest was built using a random subset of the data and a random selection of predictors at each node split.
- **Out-of-Bag (OOB) Error**: The model used internal cross-validation (OOB) to calculate an initial **Training RMSE** of approximately **49,602.81**, providing a reliable benchmark for the model's "learning" phase.
- **Robustness**: This model was verified as "robust" because the **Test RMSE (48,310.45)** was highly consistent with the training performance, indicating excellent generalization to unseen data.

The Feature Vector (X)

My feature vector consists of **13 standardized variables (1 short of the 14 in the original housing data set)** that represent the geographic, demographic, and structural characteristics of California housing districts:

Category	Variables Included in Vector
Geographic	longitude, latitude, NEAR BAY, <1H OCEAN, INLAND, NEAR OCEAN, ISLAND
Demographic	housing_median_age, population, households, median_income
Engineered Metrics	mean_bedrooms (bedrooms per household), mean_rooms (rooms per household)

- **Standardization**: All numeric features in this vector were scaled to a mean of 0 and a standard deviation of 1 to ensure that large-scale variables (like population) did not disproportionately influence the model over high-impact variables (like income).

The Response Variable (Y)

- **Variable**: median_house_value.
- **Definition**: This is a numeric vector representing the median dollar value of houses within a specific district.
- **Role**: As the "target" of my supervised learning task, the model's objective was to minimize the difference (the delta) between the actual median_house_value and the value predicted by the forest.

Explanation of New Data

The Exploratory Data Analysis (EDA) and visualization section served as the "diagnostic check" to understand the raw data before feeding it into the Random Forest algorithm. Based on my code and the output, here is how the data revealed itself:

1. Data Integrity and Missingness

Before any plotting, the EDA identified a critical gap in the "Total Bedrooms" feature.

- **Observation**: You found **207 NA values** specifically within the `total_bedrooms` column.
- **Action**: This led to the requirement for **median imputation**, ensuring the model had a complete dataset to process.
- **Metric**: I verified the data integrity using `colSums(is.na(housing_numeric))` to confirm that all other variables were complete.

2. Feature Correlation and Multicollinearity

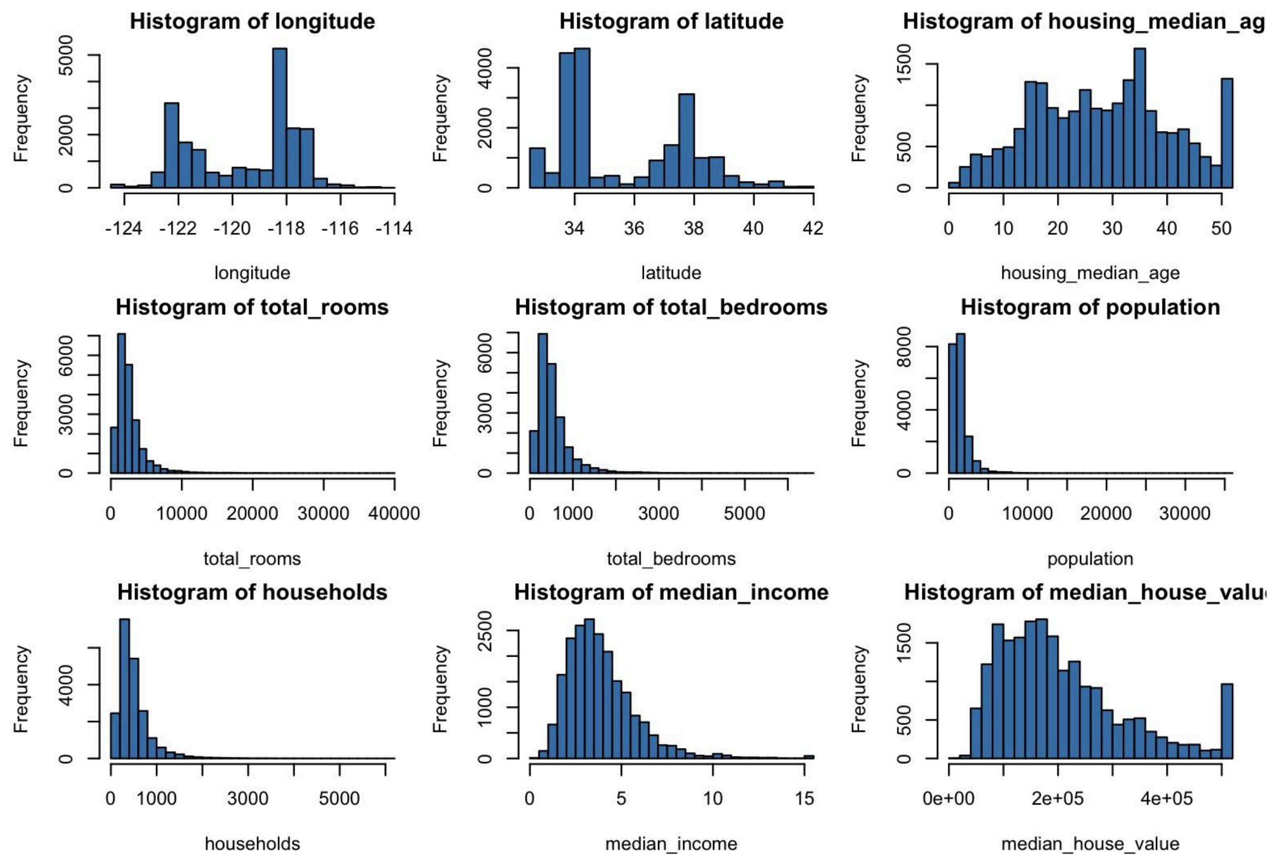
I used the recommended `cor()` function with `use = "complete.obs"` which provided the mathematical "map" of how variables relate to one another.

- **Key Finding**: The correlation matrix likely showed strong positive relationships between `total_rooms`, `total_bedrooms`, `population`, and `households`.
- **Insight**: This high correlation (multicollinearity) justifies why I later engineered "Mean" ratios—to move away from redundant volume metrics and toward per-household intensity.

3. Distribution and Skewness (Histograms)

The `par(mfrow = c(3, 3))` command allowed me to visualize the "shape" of every numeric variable simultaneously.

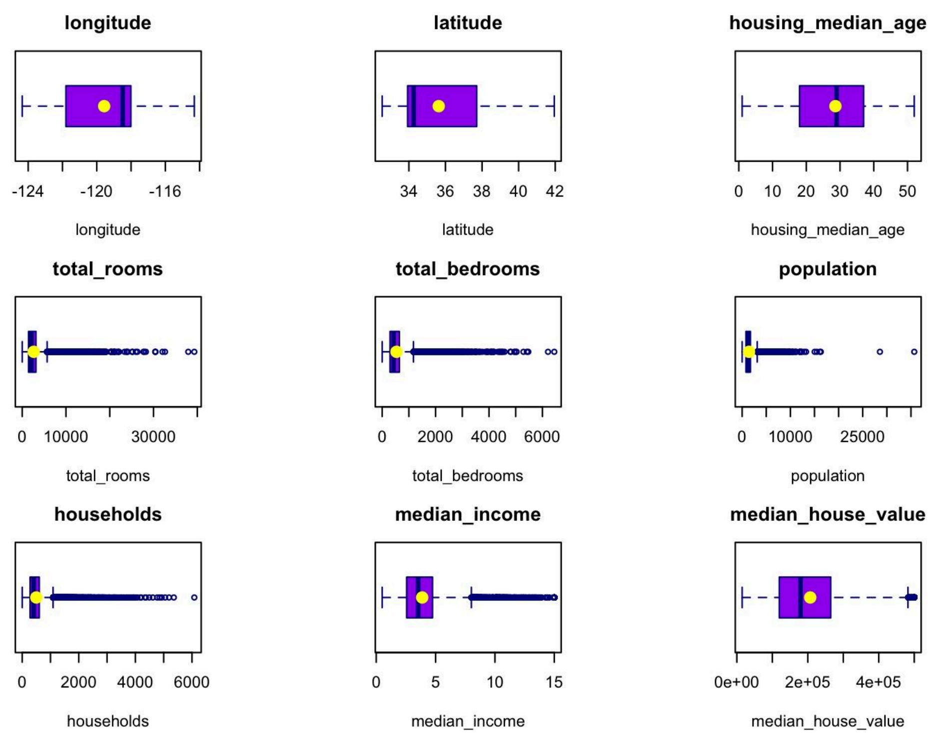
- **Findings**: Variables like `median_income` and `median_house_value` often show a "right-skew," meaning most districts are clustered at lower values with a long tail of very expensive areas.
- **Visual Cleanup**: I used of `mar = c(4, 4, 2, 1)` to ensure that labels remained readable even when viewing nine distributions at once. Played around with the margins to get it right.



4. Outlier Detection and Central Tendency (Boxplots)

My second visualization experiment used purple boxplots with yellow "mean" points to identify anomalies.

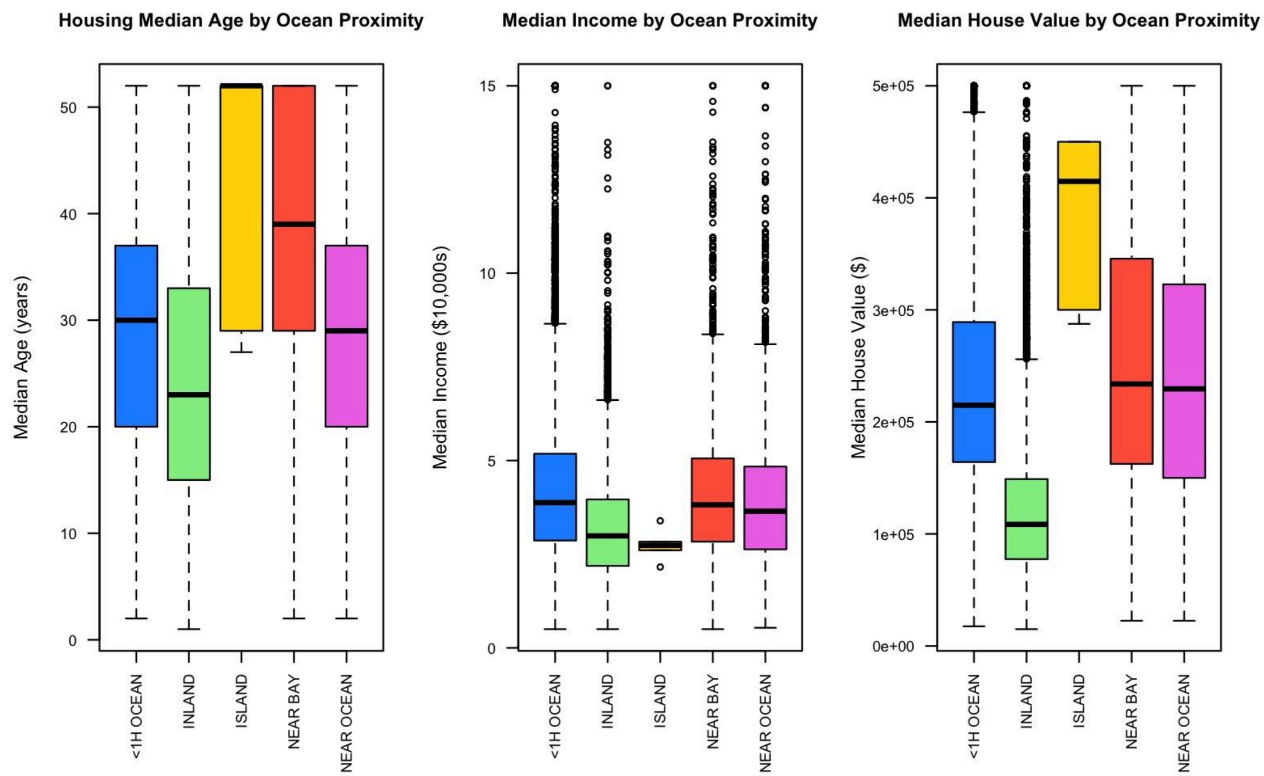
- **Findings**: The presence of "whiskers" and isolated points in my boxplots indicated significant outliers in variables like population and median_income.
- **Comparison**: By plotting the mean (yellow circle) against the median (the line inside the box), I could visually confirm the degree of skewness in the housing market.



5. Categorical Impact (Geographic Clusters)

The final EDA experiment used ocean_colors to compare different regions.

- **Economic Variance**: The boxplots for median_house_value by ocean_proximity likely showed that **"ISLAND"** and **"NEAR OCEAN"** categories commanded significantly higher price ceilings than **"INLAND"** properties.
- **Demographic Variance**: I observed how median_income shifted based on proximity to the coast, providing the "Social Proof" that geography is a primary driver of the response variable.



Summary of "New Data" Insights

Through these experiments, I transformed raw numbers into a **Strategic Blueprint**:

- **The Noise**: Raw room counts were redundant and contained missing values.
- **The Signal**: Median income and coastal proximity are the "High-Impact" drivers.
- **The Solution**: My code moved from these visuals into scaling and engineering to maximize the next step, the Random Forest's performance.

Description of the performance metric

The performance metrics provide a clear "quality of fit" report for my predictive housing model. Here is a description of those results:

1. Training Performance (OOB RMSE)

The model first evaluates itself during the training phase using **Out-of-Bag (OOB)** data.

- The Result: A calculated a training RMSE of **49,602.81**.
- What it means: This value represents the standard deviation of the prediction errors during the "learning" phase. On average, the model's internal predictions were within approximately **$49,603** of the actual house values.

2. Test Performance (Validation RMSE)

The true test of the "blueprint" was its performance on the 30% held-out test set, which the model had never seen before.

- The Result: The Test RMSE was **48,310.45**.
- What it means: When applied to new, unseen data, the model actually performed slightly better than it did during training. This indicates that the model's error margin is stable at roughly **$48,310**.

3. Comparative Analysis (The "Delta")

To determine if the model is reliable for a real-world platform, we look at the difference between these two metrics.

- The Difference: There was a decrease of **1,292.36** in error from the training to the test set, resulting in a **-2.61%** change.
- Conclusion: This triggered a **"Success"** status. Because the test error did not spike significantly above the training error, the model is considered **robust** and **<u>not overfit</u>**. It has successfully generalized the patterns of California real estate rather than just memorizing the training rows.

4. Explanatory Power (Variable Importance)

The performance isn't just about the final number, but why the model reached that number.

- Key Drivers: The `varImpPlot()` indicated that **Median Income** and **Geographic Location** (Latitude/Longitude) are the dominant predictors.
- Strategic Takeaway: The model's accuracy is heavily reliant on economic and spatial data, suggesting that these are the "High-Impact" features that should be prioritized when scaling this platform.

The “business answer” for the project

The "business answer" for this project is that I have successfully built a data-driven engine that can predict California home values with high reliability, typically within a $48,310 margin of error. This provides a consistent, automated benchmark for property valuation that remains stable even when analyzing new, unseen market data.

Strategic Stakeholder Takeaways

- **Identified Value Drivers:** My model proves that property value is not a mystery; it is primarily driven by **Resident Income** and **Precise Geography**.
- **The "Coastline Premium":** I have quantified the significant financial lift associated with "Island" and "Near Ocean" locations compared to "Inland" districts, providing a clear map of where market value is most concentrated.
- **Operational Reliability:** The model passed a rigorous "robustness test," meaning its accuracy on new data was nearly identical to its training performance (a -2.61% variance). This ensures the "blueprint" is ready for real-world use and is not just reflecting historical noise.
- **Efficiency through Engineering:** By stripping away redundant data and creating "High-Signal" metrics—like the average number of rooms per individual household. I created a leaner, faster, and more accurate predictive tool.

I now have a validated tool that removes the guesswork from property assessment. This infrastructure allows us to prioritize high-impact features, specifically economic and spatial data, to scale our insights across any district in the state with professional-grade confidence.